Basic Tukey reductions for selective and Ramsey filters on general topological Ramsey spaces

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Joint work with Natasha Dobrinen & José G. Mijares

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[1] Dobrinen *Survey on the Tukey theory of ultrafilters*

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Suppose that $\mathcal{U}$ and $\mathcal{V}$ are ultrafilters.

- A function f from $\mathcal{U}$ to $\mathcal{V}$ is **cofinal** if every cofinal subset of $(\mathcal{U}, \supseteq)$ is mapped by f to a cofinal subset of $(\mathcal{V}, \supseteq)$.

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- If $\mathcal{U} \leq_T \mathcal{V}$ and $\mathcal{V} \leq_T \mathcal{U}$ then we write $\mathcal{V} \equiv_T \mathcal{U}$ and say that $\mathcal{U}$ and $\mathcal{V}$ are **Tukey equivalent.**

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Fact

If $\mathcal{U} \leq_{RK} \mathcal{V}$ *then* $\mathcal{U} \leq_T \mathcal{V}$.

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For a given ultrafilter $\mathcal{U}$, what is the structure of the Rudin-Keisler ordering of the isomorphism classes of ultrafilters Tukey reducible to $\mathcal{U}$?

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Definition

An ultrafilter $\mathcal{U}$ on a countable base X is a **p-point** if for each $f : X \to X$ there exists $Y \in \mathcal{U}$ such that either $f \restriction Y$ is finite-to-one or $f \restriction Y$ is constant.

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Theorem (Dobrinen and Todorcevic, [3])

If $\mathcal{U}$ is a p-point ultrafilter on ω then $\mathcal{U}$ has continuous Tukey reductions.

Notation

Suppose that

- $(\mathcal{R}, \leq, r)$ is a fixed triple satisfying axioms **A.1** – **A.4**.

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- For each $X \in \mathcal{R}$ and $s \in \mathcal{AR}$,

$$\operatorname{depth}_X(s) = \begin{cases} \infty & \text{if } \nexists n \; s \leq_{\text{fin}} r_n(X) \\ \min\{n < \omega : s \leq_{\text{fin}} r_n(X)\} & \text{otherwise.} \end{cases}$$

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- If $\mathcal{C}$ is a subset of $\mathcal{R}$ we let $\mathcal{C} \geq_T \mathcal{V}$ denote the statement $(\mathcal{C}, \geq) \geq_T (\mathcal{V}, \supseteq)$.

Definition

For $X, Y \in \mathcal{R}$ and $n < \omega$, $X/n \leq Y$ means that there exists $s \in \mathcal{AR}$ such that $\mathrm{depth}_{\mathbb{A}}(s) = n$ and $[a, X] \subseteq [a, Y]$.

Selective filters for general topological Ramsey spaces

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Remark

The collection of maximal selective filters on the Ellentuck space coincides with the collection of selective ultrafilters on ω.

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5. $s \sqsubseteq t \in \mathcal{AR}$ implies that $\check{f}(s) \sqsubseteq \check{f}(t)$; and
6. $\check{f}$ is monotonic, that is, if $s, t \in \mathcal{AR}$ with $s \leq_{\mathrm{fin}} t$, then $\check{f}(s) \subseteq \check{f}(t)$.

Basic Tukey reductions and selective filters

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Let $d : \mathcal{R} \to \mathcal{P}(\omega)$ be given by

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Theorem (DMT, [2])

If $\mathcal{C}$ is a selective filter on $(\mathcal{R}, \leq)$ and $\{d(X) : X \in \mathcal{C}\}$ generates an ultrafilter on ω then $\mathcal{C}$ has basic Tukey reductions.

Fronts and Nash-Williams Filters

Definition

Suppose that $\mathcal{C}$ is a subset of $\mathcal{R}$ and $\mathcal{F}$ is a subset of $\mathcal{AR}$. $\mathcal{F}$ is a **front on** $\mathcal{C}$ if

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Definition

- A subset $\mathcal{C} \subseteq \mathcal{R}$ **satisfies the Abstract Nash-Williams Theorem** if for each family $\mathcal{G} \subseteq \mathcal{AR}$ and partition $\mathcal{G}_0 \cup \mathcal{G}_1 = \mathcal{G}$, there is a $C \in \mathcal{C}$ and $i < 2$ such that $\mathcal{G}_i|C = \emptyset$.
- A subset $\mathcal{C} \subseteq \mathcal{R}$ is a **Nash-Williams filter on** $(\mathcal{R}, \leq)$ if it is a filter on $(\mathcal{R}, \leq)$ that satisfies the Abstract Nash-Williams Theorem.

Nash-Williams Filters & Basic Tukey Reductions

Proposition (DMT, [2])

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Theorem (DMT, [2])

Suppose $\mathcal{C} \subseteq \mathcal{R}$ is a Nash-Williams filter on $(\mathcal{R}, \leq)$ and $\{d(X) : X \in \mathcal{C}\}$ generates an ultrafilter on ω. Then an ultrafilter $\mathcal{V}$ on ω is Tukey reducible to $\mathcal{C}$ if and only if $\mathcal{V} = f(\langle \mathcal{C} \restriction \mathcal{F} \rangle)$ for some front $\mathcal{F}$ on $\mathcal{C}$ and some function $f : \mathcal{F} \to \omega$.

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It is consistent with ZFC that the following statements hold.

- *Every finite Boolean algebra appears as the initial Tukey structure below some p-point.*
- *The structure of the Tukey types of p-points contains the partial order* $([\omega]^{<\omega}, \subseteq)$ *as an initial structure.*

Applications

Theorem (DMT, [2])

Let $\mathcal{K}$ be a Fraïssé class of finite ordered relational structures satisfying the Ramsey property. There is a topological Ramsey space $\mathcal{R}$ such that if $\mathcal{C}$ is a Nash-Williams filter on $(\mathcal{R}, \leq)$ then the Rudin-Keisler ordering of the p-points Tukey reducible to $\mathcal{C}$ is isomorphic to the partial order $(\tilde{\mathcal{K}}, \leq)$.

Applications

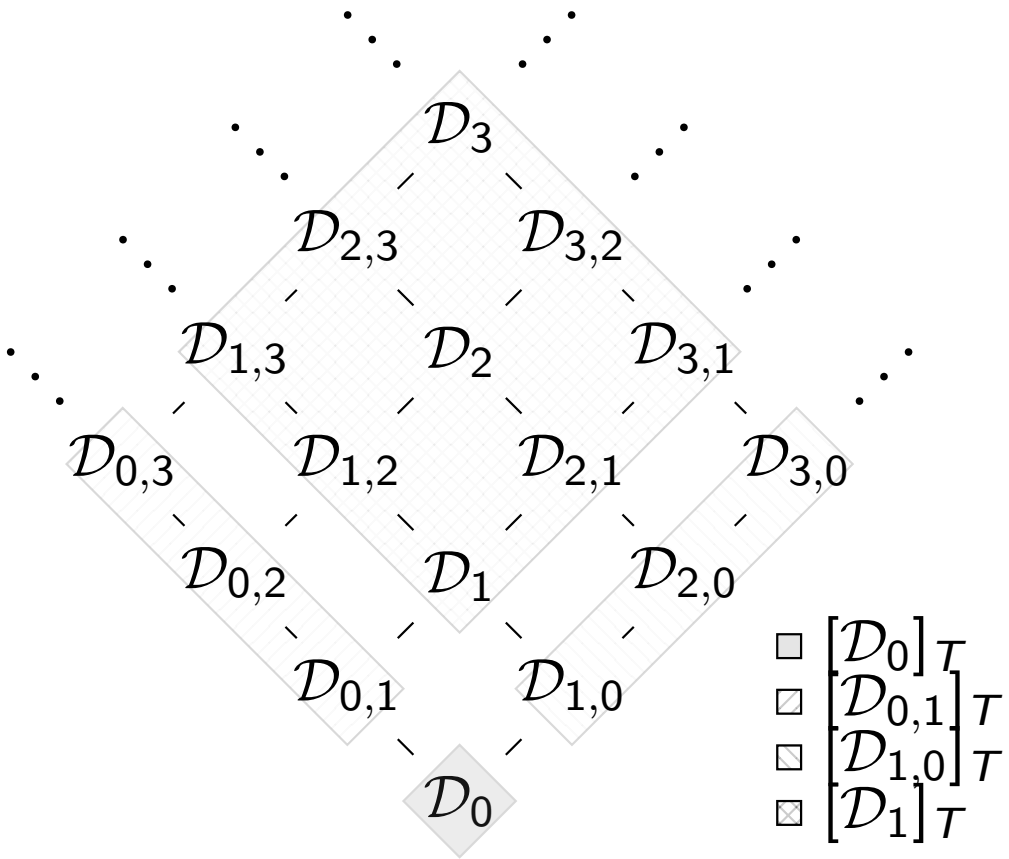

Figure: Rudin-Keisler structure of the p-point ultrafilters within the Tukey types of nonprincipal ultrafilters Tukey reducible to $\mathcal{D}_1$

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Thank You!

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Survey on the Tukey theory of ultrafilters.
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